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Privacy in an augmented reality

Andreas Kotsios*

ABSTRACT

Google introduced Google Glass in 2012. Even though this wearable augmented reality device has only been introduced to the public in some kind of testing (beta) mode, it has managed to draw the attention of the public, partially because of the possibilities that such a device can provide and partially because of the dystopian scenarios regarding privacy. This article seeks to identify the challenges related to privacy arising from the use of this type of devices. Furthermore, it examines the existing legal framework with regards to protection of privacy in the two main markets of these devices, namely the USA and EU, and recommends some solutions that could be used as a starting point to augment the protection of privacy and at the same time give space to the new technologies to be further developed.

KEYWORDS: privacy, wearables, augmented reality, Google Glass

1. INTRODUCTION

‘The most profound technologies are those that disappear. They weave themselves into the fabric of everyday life until they are indistinguishable from it.’¹

People always try to find ways to communicate information in the most effective way, namely in a way so that the information is perceived without external distractions, such as the distractions created by the medium that bears the information.² As an example we can think of street signs: we create them and place them in a way that can be easily seen and become understood. In this way the information is communicated without further thinking and we can focus on the meaning of the information, for example, the word ‘STOP’, and not on the medium of the information, namely the metal plate that the word ‘STOP’ is written on.

Until now, the information that was communicated through computers was communicated with the distraction created by the computer, since the medium, namely the computer, stood between the information and the receiver of the information, that is, the user of the device. The user of the computer would find the information that he/she requested after writing, double clicking and generally focusing on the device, the medium of the information, instead of focusing directly just on the information. In the last few years though, some new technologies are trying to change

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1 Mark Weiser, ‘The Computer for the 21st Century’ (1991) 265(3) *Scientific American* 94.

2 See ‘Semiotics’ (2014), Wikipedia, the free encyclopedia <<http://en.wikipedia.org/w/index.php?title=Semiotics&oldid=586750776>> accessed 11 January 2014.

this picture. These technologies are augmented reality (AR) and wearable devices. The wearable devices with the power of AR technology are trying to simplify and make more effective our everyday life by minimizing the interference between man and information.

Even though these technologies are not so new—Steve Mann created his first wearable AR device, Eye Tap, in the 1980s³—it is not until 2012 that they have become that popular. That is when Google entered the market of wearable AR devices by the creation of Google Glass.⁴ Despite the fact that Google Glass, until now, has been introduced to the public only in beta (test) mode the impact and the buzz about this device is enormous. Many are fascinated by the device and have already found lucrative and productive ways to use it, something which will be presented later on in this article. According to research by Business Insider Intelligence, the expected unit sales for Google Glass will be 21 million units annually by year-end 2018.⁵

Many persons worldwide, however, are very concerned about the legal and social implications that these new devices may raise. There are already people talking about using violence against users of Google Glass.⁶ The main reason is that people feel that their privacy is intruded by this new technology,⁷ since their image and other information can be captured by these devices without their consent—many times without them even noticing that their information is captured.

This article will focus on the protection of the privacy of the third person from individuals/users of the devices and, to some extent, from non-governmental entities, such as the companies that produce the devices and provide the operating systems and applications as well as associated third parties.

The first section of the article examines the AR and wearable technology and also their present and possible future uses. Section 2 examines first the reactions against these technologies; then the article will present a taxonomy of privacy to use it for the identification of the privacy issues that arise by the use of wearable AR devices. Section 3 examines the law that relates to privacy protection; what legal instruments can be used by the third persons in case of privacy intrusion; the US and the EU approach. It will also examine if the existing protection is adequate and effective as well as what are the loopholes that exist. The last section of the article will present some

3 'EyeTap' (2014), Wikipedia, the free encyclopedia <<http://en.wikipedia.org/w/index.php?title=EyeTap&oldid=582504369>> accessed 16 January 2014.

4 'Google Glass' <<http://www.google.com/glass/start/>> accessed 4 December 2013.

5 'BI INTELLIGENCE FORECAST: Google Glass Will Be An \$11 Billion Market By 2018' (2013), Business Insider <<http://www.businessinsider.com/google-glass-11-billion-market-by-2018-2013-5>> accessed 4 December 2013.

6 For example, 'Meetup.com CEO: I Want to Punch Google Glass Wearers in the Face' (14 May 2013), The Local Voluntary <<http://www.localvoluntary.com/2013/05/meetup-com-ceo-i-want-to-punch-google-glass-wearers-in-the-face/>> accessed 16 January 2014.

7 See 'The Human Cloud: Wearable Technology from Novelty to Production, a Social Study into the Impact of Wearable Technology' (2013), Centre for Creative and Social Technologies (CAST), Goldsmiths College, University of London, Rackspace Hosting <http://www.rackspace.co.uk/sites/default/files/whitepapers/The_Human_Cloud_-_June_2013.pdf> accessed 7 January 2015, where it is cited that 62 per cent of the respondents think that Google Glass should be regulated and 20 per cent that these devices should be banned entirely.

recommendations to solutions that could fill the existing loopholes and create an *a priori* protection of privacy without restricting these technologies.

2. DEVICES AND TECHNOLOGY

2.1 Augmented reality

Today, if a person wants to find a piece of information, he/she will go to a PC, sit down, turn it on or 'wake it up' and then start browsing and searching in different web pages to find the information that he/she is looking for. Alternatively, he/she will take her mobile phone out of his/her pocket or pick up a tablet, unlock it, open the browsing application or maybe another search application, and then start searching by trying different web pages to find the information that he/she needs.

The problem with both these ways is that even though the information a person tries to find is many times connected to the physical world, the means to find the information is a digital device. The information is cut from the physical object that the user wants to find and this leads to a gap in communication. That means that the communication of the information is not effective like the example of the street sign that was mentioned above, since between the information and the person who seeks the information you have the 'noise' of the device.

AR is the technology that is trying to erase that 'noise'. The goal of this technology is to communicate the information directly by making the device that uses the AR 'disappear'. That happens by superimposing the information of the physical object up on the object. As Steve Mann, one of the pioneers of wearable AR devices, stated, '[a]ugmented reality means to super-impose an extra layer on the real-world environment, thereby augmenting it. An "augmented reality" is thus a view of physical, real-world environment whose elements are augmented by computer-generated sensory input such as sound, video, graphics or GPS data.'⁸

AR is part of the general concept of mediated reality, which not only adds to reality but also can diminish or modify it. As an example, a device in the shape of eyeglasses using mediated reality can make the wearer see through different spectral bands, such as infrared, or even download and update continuously the eyeglass prescription that is suitable for the eyes of the user of the device.⁹ Even though AR is part of mediated reality, the term 'augmented reality' is used as a commercial term to express any augmentation, diminishing or modification of reality by computer-generated sensory input.

2.2 Wearable devices

To augment reality, a device combines the digital and physical data as an input to create a unified output. Most modern computers that have a camera and a powerful processor can achieve this, but the problem is that the user is still distracted by the device, such as the pc, the mobile phone or the tablet. Specifically, the user has to unlock the device, try to find the right application, choose the right application and

8 Steve Mann, 'Wearable Computing' in Soegaard Mads and Dam Rikke Friis (eds), *The Encyclopedia of Human-Computer Interaction* (2nd edn, The Interaction Design Foundation, 2013) <http://www.interaction-design.org/encyclopedia/wearable_computing.html> accessed 4 December 2013.

9 *ibid.*

then point the device to the right direction in order for the camera of the device to 'catch' the physical object for which the information was requested. Furthermore, instead of the physical object becoming the main object where the unified information is projected, the unified information is projected on the screen of the device, which means that instead of looking at the physical object for which the user seeks the information, the user looks at the screen of a device. Therefore, even though the information is communicated in relation to the physical object there is still the 'noise' of the device between the information and the person seeking it. This is possibly one of the reasons why AR applications did not manage to get a big share of the market until now.

The wearable AR devices, however, are here to solve this problem. They can be worn by the user and incorporate a computer, input/output (I/O) components, such as cameras and audio components, GPS sensors and other components. Until now most of these devices were used either by the army¹⁰ or in research.¹¹ Early in 2012, though, a new head-worn AR device came out and is expected to make a real difference in wearable AR devices. The device was created by the IT giant Google and is called Google Glass.¹²

Google Glass, by the time of the publication of this article, has been launched only as a beta edition, or as Google names it 'the Glass Explorer Program'—only some people are allowed to buy the device to use it and give feedback to Google before the device goes public.¹³ Google Glass is a head-mounted wearable AR device that can display information on a transparent prism, which is positioned in front of the right eye of the user, thus minimizing the interference between information and the user of the device, since the user is not distracted by the device. Therefore, the device can augment reality by adding a layer of information on the physical world. Google Glass also has a camera at eye level of the user thus making it possible to take photos and videos easily and seamlessly. It has, moreover, the ability to use face recognition applications to identify individuals in real time,¹⁴ even though for the time being Google has limited this possibility.¹⁵

10 For example, 'Helmet-Mounted Display' (2013), Wikipedia, the free encyclopedia <http://en.wikipedia.org/w/index.php?title=Helmet-mounted_display&oldid=582351291> accessed 4 December 2013, where it is stated that these devices are used to allow cuing of weapon systems.

11 See n 3.

12 See n 4.

13 See Drew Olanoff, 'Google Says All 2,000 Glass Explorers Have Been Invited To Pick Up Their Device' (2013), Tech Crunch <<http://techcrunch.com/2013/05/16/google-has-now-invited-all-2000-people-who-signed-up-at-last-years-io-to-pick-up-glass/>> accessed 13 December 2013, where it is stated that 2,000 developers and 8,000 non-developers got their glasses to use them and give feedback to Google. It should be noted here that on the 19th of January 2015 Google announced that they terminate the Explorer Program to prepare 'the future version of Glass'; see Google Glass, 'We're graduating from Google[x] labs', (Post shared publicly on Google+, 15 January 2015) <<https://plus.google.com/±GoogleGlass/posts>> accessed 20 January 2015.

14 For example, 'Google Glass Real-Time Facial Recognition Arrives with "NameTag"' (2013), Slash Gear <<http://www.slashgear.com/google-glass-real-time-facial-recognition-arrives-with-nametag-19309472/>> accessed 11 January 2014.

15 Charles Arthur, 'Google "Bans" Facial Recognition on Google Glass - but Developers Persist' *The Guardian* (3 June 2013), London, United Kingdom <<http://www.theguardian.com/technology/2013/jun/03/google-glass-facial-recognition-ban>> accessed 16 January 2014.

The applications that presently exist are limited to approximately 100, official and unofficial,¹⁶ and therefore many AR possibilities are still to be expected when more applications will be developed. Even though no one really knows if Google Glass will manage to be the success that Google wants, it has definitely brought many social and legal issues to the surface, issues that were either overlooked until now or triggered by the use of this device. Furthermore, this device has shown the public the uses and possibilities of AR technology for the first time.

2.3 Uses of wearable AR devices

Wearable AR devices are here to make our lives easier and even more interesting, and can be used in a vast amount of different situations. The range of the applications of this technology is limitless. They stretch from simple fun, such as games¹⁷ and sports,¹⁸ to navigation,¹⁹ tourism²⁰ and advertisement.²¹ Even more important are the medical applications of this technology. Some are already being developed such as, for example, an application to be used by surgeons for monitoring vital signs of a patient during an operation.²² The same application can be used to seek remote

- 16 'Google Glass Application List' <<http://glass-apps.org/google-glass-application-list>> accessed 13 October 2014.
- 17 See 'Ingress' <<http://www.ingress.com/>> accessed 4 December 2013. An AR game which has already been launched by Google, where the gamer must choose one of the two teams to control 'a mysterious energy' that some scientists 'unearthed'. The gamer has to move in real environment with the help of her android device and interact with other people in real world, and not in virtual world, as it was the case until now in computer games, to discover the sources of this mysterious energy. Some interesting information about the game can be found in 'The Demographics of Ingress' (23 January 2013), Simulacrum <<http://simulacrum.cc/2013/01/23/the-demographics-of-ingress/>> accessed 13 December 2013. Here it should be added that even though ingress is still not available for Google Glass, from the beginning the production team was focusing in the wearable devices. As Brandon Badger, the product manager for Ingress at Niantic Labs, said in an interview, '[w]e had all of this great geographic data and devices coming down the pipeline like Glass and wearable computing. So we saw an opportunity to build more interesting user experiences in the gaming realm. The idea behind Ingress was to get people off the couch and moving in the streets.' Parts of the interview can be found in Dean Takahashi 'Google Niantic's Ingress augmented-reality game grows with real-time events', Venture Beat, (2013) <<http://venturebeat.com/2013/10/15/google-niantics-ingress-augmented-reality-game-grows-with-real-time-events/>> accessed 4 December 2013.
- 18 For example, 'Instabeat: Master Your Swim!' <<http://www.instabeat.me/>> accessed 4 December 2013 and 'Play Golf Using Google Glass with New GolfSight GPS Rangefinder' (2013), Digital Journal <<http://www.digitaljournal.com/pr/1596621>> accessed 13 December 2013.
- 19 For example, 'Mercedes-Benz to Offer Google Glass Navigation App' (2013), CarAdvice <<http://www.caradvice.com.au/257314/mercedes-benz-to-offer-google-glass-navigation-app/>> accessed 4 December 2013.
- 20 For example, 'Field Trip' <<http://www.fieldtripper.com/>> accessed 4 December 2013.
- 21 For example, 'United States Patent: 8510166 - Gaze Tracking System', USPTO, (2013) <<http://patft.uspto.gov/netacgi/nph-Parser?Sect1=PTO2&Sect2=HITOFF&u=%2Fnetacgi/html%2FPTO%2Fsearch-adv.htm&r=36&p=1&f=G&l=50&d=PTXT&S1=%2820130813.PD.+AND+Google.ASNM.%29&-OS=ISD/20130813+AND+AN/Google&RS=%28ISD/20130813+AND+AN/Google%29>> accessed 4 December 2013. The patent is about a system that could be used in wearable AR devices to give information about what the user of the device is gazing.
- 22 'Philips Demonstrates Medical Applications of Google Glass' (2013), The Engineer <<http://www.theengineer.co.uk/medical-and-healthcare/news/philips-demonstrates-medical-applications-of-google-glass/1017229.article>> accessed 4 December 2013.

assistance.²³ Additionally, Google Glass has already been used as a method of co-operation between doctors. Specifically, on 12 September 2013 an orthopaedic surgeon, Ponce, performed a shoulder replacement at a hospital in Birmingham, and another surgeon, Dantuluri, from Atlanta was able to watch and most importantly to interact by using Google Glass. According to the article, '[t]he built-in camera transmitted the image of the surgical field to Dantuluri. VIPAAR allowed Dantuluri, who saw on his computer monitor exactly what Ponce saw in the operating room, to introduce his hands into the virtual surgical field. Ponce saw Danturuli's hands as a ghostly image in his heads-up display.'²⁴

Furthermore, wearable AR devices are giving people with disabilities in mobility, vision and hearing the opportunity to get a better quality of life. For instance, the development of speech recognition will help deaf people understand what their interlocutor is saying to them by displaying in real time the words on the 'prism'.²⁵ Also, emotional recognition could help people with Asperger syndrome.²⁶ Even more, face and spatial recognition could aid blind people in their everyday life by announcing the people and the buildings that the blind person is pointing to.²⁷ Paraplegics can use devices like this to take photos or just browse through voice dictation.²⁸ The possibilities are just limitless; and this is just the beginning.

Additionally, according to a Juniper report, there are already 60 million people that use AR apps on a regular basis, a number that will increase to 200 million by 2018.²⁹ All industries are ready to invest in the field of AR and since the optimization of this technology can be realized best through the use of wearable devices, it can be expected with certainty that the wearable AR devices are going to be the dominant technology in the coming years.³⁰

All the possibilities that are briefly mentioned here arose within the last two to three years. Many of them arose just after 2012 when Google Glass started the 'Explorer Program'. If we add to this picture the assumption that, according to Moore's law,³¹ the processing power of computers doubles every 18–24 months, we can easily imagine the changes that wearable AR devices will bring.

23 *ibid.*

24 'First Virtual Surgery with Google Glass' (2013), ScienceDaily <<http://www.sciencedaily.com/releases/2013/11/131108140403.htm>> accessed 4 December 2013.

25 For what the prism is see 'Google Glass (infographic) - How It Works' <<http://www.brillen-sehhilfen.de/en/googleglass/>> accessed 13 December 2013; Marco della Cava, 'Beyond a Gadget: Google Glass is a Boon to Disabled' *USA Today* (23 October 2013), United States of America <<http://www.usatoday.com/story/tech/2013/10/22/google-glass-aids-disabled/3006827/>> accessed 17 January 2014.

26 Della Cava, *ibid.*

27 *ibid.*

28 *ibid.*

29 The report can be found in 'Mobile Augmented Reality Smartphones, Tablets and Smart Glasses 2013-2018' (2013), Juniper Research <http://www.juniperresearch.com/reports/mobile_augmented_reality> accessed 4 December 2013. See also 'Smartphones + Smart Glasses: Augmented Reality to Jump 333% by 2018' (2013), Venture Beat <<http://venturebeat.com/2013/11/06/smartphones-smart-glasses-augmented-reality-to-jump-333-by-2018/>> accessed 4 December 2013.

30 See also 'HoloLens' <<http://www.microsoft.com/microsoft-hololens/en-us>> accessed 30 January 2015, where another IT giant, Microsoft, presents its new wearable AR device.

31 'Moore's Law' (2013), Wikipedia, the free encyclopedia <http://en.wikipedia.org/w/index.php?title=Moore%27s_law&oldid=585838794> accessed 27 December 2013. I have to point out here that Moore's Law, according to many scientists, is not working anymore. That does not mean

Like most radical and life changing technologies, wearable AR devices have already brought to light legal and social issues and challenges. The next section of this article will look at these issues.

3. THE ISSUES

Even though Google Glass, as already stated above, is not the first wearable AR device created and is not even properly out on the market yet, great discussions are already taking place regarding the social and legal problems that wearable AR devices create.

Intellectual property issues related to wearable AR devices are definitely going to be discussed in the near future, such as the question whether the augmented reproduction of a physical piece of art constitutes distortion or modification.

But the biggest part of the reactions to Google Glass is in connection with questions on privacy. Privacy concerns were raised both for the user of the device and also for third persons who are part of the environment of the device, namely persons whose image can be captured and therefore digitized by the wearable AR device, thus leading to the processing of their information.

This section of the article will start by presenting reactions related to privacy brought on by this new technology. To see which of the issues are really connected to privacy, the article will try to find the notion of privacy and present the existing privacy protection legislation in the EU and the USA, which are the main markets that Google Glass and other similar devices are expected to target primarily.

Furthermore, the article will focus only on the protection of privacy of third persons/non-users of the wearable AR device from individuals/users of the wearable AR device and other private entities.

3.1 The reactions

The reactions of the press, governments and the general public were mixed when Google Glass was released. There were those who praised the device and the potentials that a commercial wearable AR device could bring. *Time* Magazine included the device in the list of the 'Best Inventions of the Year 2012'.³² In addition, the anticipation of people can be seen in the bids of one of the first Google Glass that were offered on eBay: the device, which cost normally \$1500 managed to get bids of up to \$95,300 in only four days, before the seller closed the auction since this was against the terms and conditions of Google.³³ As the 'Explorer Program' started by using an

though that computing performance will stay stable. For more information, see presentation by Robert Colwell, Director of DARPA, in Hot Chips Symposium in August 2013 'The Chip Design Game at the End of Moore's Law' <http://www.hotchips.org/wp-content/uploads/hc_archives/hc25/Hc25.15-key-note1-Chipdesign-epub/Hc25.26.190-Keynote1-ChipDesignGame-Colwell-DARPA.pdf> accessed 16 January 2014.

32 'Best Inventions of the Year 2012' (2012), *Time* Tech <<http://techland.time.com/2012/11/01/best-inventions-of-the-year-2012/slide/google-glass/>> accessed 5 December 2013.

33 See Kashmir Hill, 'Google Glass Bid Up To \$95,300 On eBay Before Seller Realized He Isn't Allowed To Sell Them' *Forbes* (17 April 2013), United States of America <<http://www.forbes.com/sites/kashmir-hill/2013/04/17/google-glass-bid-up-to-95300-on-ebay-before-seller-realized-he-couldnt-sell-them/>> accessed 9 December 2013.

invite system for purchasing Google Glass, there were invitations on eBay that paid up to \$5000, and although these bids were possibly scams,³⁴ there was definitely a big momentum at that time for Google Glass.

The reactions against Google Glass were also immense. Movements against Google Glass, such as Stop The Cyborgs,³⁵ arose as soon as Google Glass appeared. Some businesses, such as bars and casinos, have already banned the use of Google Glass on their premises.³⁶ The press has also contributed in many instances to the spreading of fear about the dystopian future that the wearable devices will bring.³⁷ A survey in the UK showed that 20 per cent of people thought that Google Glass and other wearable technologies should be banned.³⁸ These reactions are connected mostly to the ability of the device to take photos and videos seamlessly and without people noticing it.

In addition, in the USA, members of the Bipartisan Privacy Caucus sent a letter to Google CEO Larry Page asking several questions related to privacy.³⁹ Furthermore, data protection authorities from a number of countries have sent a letter to Google expressing privacy concerns related to the use of Google Glass, such as how is Google going to address issues related to the information gathered by the device, face recognition, security, etc.⁴⁰

As we can see, there is a general concern related to privacy and data collection connected to videotaping and photographing with such devices as well as to face recognition possibilities that these devices provide.

34 See Jordan Valinsky, 'You Can Purchase Google Glass Invitations on eBay For One Dollar or \$5,000' (2013), BetaBeat <<http://betabeat.com/2013/11/you-can-purchase-google-glass-invitations-on-ebay-for-one-dollar-or-5000/>> accessed 14 December 2013.

35 'Stop The Cyborgs' <<http://stopthecyborgs.org/>> accessed 5 December 2013. As they state on their website '[t]he issue is not covert recording. Spy cameras exist and the current generation of Google Glass is not particularly good for covert recording. Rather the first issue is that that wearable devices socially normalise ubiquitous surveillance. That is they create a society where we *expect* to be recorded, where every moment to is shared, documented and data-mined. The second issue is that wearable technology blurs the line between people and tech, between individuals and institutions. The moral agency of the individual is reduced when a corporation forms part of their extended body, when every decision is mediated and influenced. The freedom of society and the possibility of genuine authentic response is threatened when every action is monitored and gamified.'

36 See Richard Gray, 'The Places Where Google Glass Is Banned' (2013), Telegraph <<http://www.telegraph.co.uk/technology/google/10494231/The-places-where-Google-Glass-is-banned.html>> accessed 5 December 2013.

37 For example, Adam L Penenberg, 'Google Glass in 10 Years: The View from Dystopia' (2013), Pando <<http://pando.com/2013/05/25/google-glass-in-10-years-the-view-from-dystopia/>> accessed 14 December 2013.

38 See n 7.

39 The letter has been taken down from the official webpage. A copy can be found in the article Meghan Kelly 'Google Glass Faces Hailstorm of Privacy Questions from U.S. Government', Venture Beat (2013) <<http://venturebeat.com/2013/05/16/google-glass-privacy-congress/>> accessed 14 December 2013.

40 'News Release: Data Protection Authorities Urge Google to Address Google Glass Concerns - June 18, 2013' (June 2013), Office of the Privacy Commissioner of Canada <http://www.priv.gc.ca/media/nr-c/2013/nr-c_130618_e.asp> accessed 5 December 2013.

3.2 Similar reactions in the past

The truth is that this is not the first time that a new technology becomes so controversial. There are various examples of technologies in modern history that created implications with regards to privacy.

The most famous example is the one of Kodak cameras, which introduced the instantaneous photographs in 1888. When the Kodak cameras first came out, they became a big success in spite of their high price, and the reason was the simplicity of the camera. As the slogan said: '[y]ou press the button, we do the rest.'⁴¹ There were other cameras before Kodak, but Kodak made it possible to get snapshots anywhere. Of course, side effects of this technology appeared instantly. The so-called 'camera fiends' started appearing on beaches taking photos of unaware female bathers.⁴² It was the first time that it became possible for people to be photographed without their consent.⁴³ Furthermore, people got distressed because the new device could capture spontaneous moment and emotions, something which was not socially acceptable at that time. This technology is the one that made Warren and Brandeis declare the famous 'right to be let alone',⁴⁴ which is considered the cornerstone of the legal conceptualization of privacy.

The reactions to such technologies, mainstream now but radical when invented, are a lot like the reactions to Google Glass and other wearable AR devices that we witness right now; many people remain sceptical even though they are fascinated by the new technology.

There are definitely privacy issues arising out of the use of wearable AR devices. Even Google, by the voice of one of its spokesmen, Courtney Hohné, has admitted that there are privacy concerns.⁴⁵ The questions arising are whether these problems are as big as some people believe, if the problems are really new and if the traditional approaches of privacy can be applicable in the world of ubiquitous computing.⁴⁶

To answer these questions the notion of privacy will be examined.

3.3 The notion of privacy

Legal conceptualization of privacy is very difficult. As Henderson perspicaciously wrote, '[p]erhaps privacy is essentially contested – most everyone agrees that we should have it, but has different ideas of just what it is.'⁴⁷ Many attempts have been

41 David Lindsay, 'The Kodak Camera Starts a Craze', Public Broadcasting Service <<http://www.pbs.org/wgbh/amex/eastman/peoplevents/pande13.html>> accessed 6 December 2013.

42 *ibid.*

43 See Samuel D Warren and Louis D Brandeis, 'The Right to Privacy' (1890) 4(5) *Harvard Law Review* 193, 195f, 7, where Warren and Brandeis are referring to the photographs of the actress Marion Manola that were taken without her consent while she was performing in tight clothes.

44 Warren and Brandeis, *ibid.*

45 David Streitfeld, 'Google Glass Picks Up Early Signal: Keep Out' *The New York Times* (6 May 2013), New York, USA <<http://www.nytimes.com/2013/05/07/technology/personaltech/google-glass-picks-up-early-signal-keep-out.html>> accessed 9 December 2013.

46 'Ubiquitous Computing' (2013), Wikipedia, the free encyclopedia <http://en.wikipedia.org/w/index.php?title=Ubiquitous_computing&oldid=583309754> accessed 5 December 2013.

47 Stephen E Henderson, 'Expectations of Privacy in Social Media' (2012) 31(symposium edition) *Mississippi College Law Review* 232.

made to find a definition of privacy—among others the right to be let alone⁴⁸ and contextual integrity⁴⁹—but as many scholars have argued the notion of privacy remains ‘elusive’,⁵⁰ ‘slippery’,⁵¹ ‘with no common denominator’⁵² and attempts to define privacy are still unsuccessful.⁵³ A reason for that is that the notion of privacy is not just a single-layer legal concept but a multi-layer concept with multiple aspects related to law, social and economic environments and politics.

Even though the goal of an unambiguous definition of privacy looks more like a chimera and, therefore, definitions that try to give a global meaning to the term privacy have difficulty in being accepted, there are legal tools which can help identify ‘problems, intrusions, categories’ or types of privacy⁵⁴ to protect it. These tools are the taxonomies of privacy. The advantage of using taxonomies instead of clear descriptions of what privacy means is that instead of trying to find the true meaning or archetype of the idea of privacy, we can categorize the elements of privacy that are worth to be protected to create a framework of protection.

Many legal scholars have created privacy taxonomies: Solove created a taxonomy of privacy problems related to (i) information collection, (ii) information processing, (iii) information dissemination and (iv) invasion.⁵⁵ Kaspar categorized privacy invasions in (i) extraction, (ii) observation and (iii) intrusion.⁵⁶ The taxonomy that will be used in this article to detect which privacy issues can arise for third persons by the use of wearable AR devices is the revised Clarke’s taxonomy⁵⁷ by Finn, Wright and Friedewald.⁵⁸ The reason is that this new taxonomy includes the human-centric approach of Clarke’s taxonomy, namely, which ‘elements of privacy are important and must be protected’, and it also takes into consideration the latest developments on new technologies that have created additional privacy issues.

According to this taxonomy, there are seven types of privacy that need to be protected:

- i. privacy of the person, which includes the protection of body functions and characteristics, among others biometrics;
- ii. privacy of behaviour and action, which includes the ability to act in private, semi-public and public spaces without being monitored and controlled;

48 Warren and Brandeis (n 43).

49 Helen Nissenbaum, *Privacy in Context: Technology, Policy and the Integrity of Social Life* (Stanford UP 2010).

50 Serge Gutwirth, *Privacy and the Information Age* (Rowman & Littlefield 2002) 30.

51 James Q Whitman, ‘The Two Western Cultures of Privacy: Dignity Versus Liberty’ (2004) 113 *The Yale Law Journal*, 1153.

52 Daniel Solove, *Understanding Privacy* (Harvard UP 2008) 12.

53 Colin J Bennett, *Regulating Privacy: Data Protection and Public Policy in Europe and the United States* (Cornell UP 1992).

54 Rachel L Finn, David Wright and Michael Friedewald, ‘Seven Types of Privacy’ In *European Data Protection: Coming of Age*, ed. S. Gutwirth et al. (Springer 2013) 3.

55 Solove (n 52) 9.

56 Debbie VS Kaspar, ‘The Evolution (or Devolution) of Privacy’ (2005) 20(1) *Sociological Forum*, 76.

57 Roger Clarke, ‘Introduction to Dataveillance and Information Privacy, and Definitions of Terms’ (1999), Roger Clarke’s Dataveillance and Information Privacy Pages, where he categorizes privacy to privacy of (i) the person, (ii) personal data, (iii) personal behaviour and (iv) personal communication.

58 Finn (n 54).

- iii. privacy of communication, namely the right to communicate without any kind of interference;
- iv. privacy of data and image, which means that individuals have control over their data, including the data from the capture of their image, and that their data cannot be made available to third parties without the individuals' knowledge;
- v. privacy of thoughts and feelings, which includes the right to keep or share their thoughts and feelings at will;
- vi. privacy of location and space, namely that individuals are not tracked, monitored or identified in public, semi-public or private space; and
- vii. privacy of association, which refers to the right of individuals to associate with whom they want without being monitored.⁵⁹

The reason why these elements of privacy have to be protected is connected to socio-economic and political aspects, such as the strengthening of democracy, the freedom of speech, the free choice of sexual orientation and religion. By protecting these types of privacy individuals/citizens will not have the fear of constant surveillance and society can benefit by the free and unbiased development of the individual.

By having in mind the typology of privacy that was just described, the next section of the article examines the possible privacy implications that arise out of the use of wearable AR devices.

3.4 Privacy implications by the use of wearable AR technology

Users of wearable AR devices can accomplish three actions that could lead to intrusions into the privacy of third persons, as will be clarified hereinafter. The three actions are: (i) the capturing of photos and videos of third persons; (ii) the uploading of these records to private, semi-public or public spaces, such as to a social network site (SNS) like Facebook, Google+ or YouTube⁶⁰; and (iii) the processing of the face and body of that person by the use of biometrics.

These actions can take place simultaneously. As an example, a user of a wearable AR device takes a video of a third person; a face recognition application is used to identify this person. At the same time the data of the location where the video was recorded are being registered in the video and after finishing the video the user of the wearable AR device is uploading it on YouTube making it accessible for everyone, everywhere in the world.

The wearable AR devices are not the first devices to introduce the above-mentioned actions, there are already devices that can capture video and photos and upload these images and videos on the Internet, there are also already applications that can proceed to face recognition. There is, though, a quantitative and qualitative change that ignites the debate about the need to protect privacy against the use of

⁵⁹ *ibid* 4–6.

⁶⁰ See also Nicole B Ellison 'Social network sites: Definition, history, and scholarship' (2007) 13(1) *Journal of Computer-Mediated Communication* 210, where SNSs are defined as 'web-based services that allow individuals to (1) construct a public or semi-public profile within a bounded system, (2) articulate a list of other users with whom they share a connection, and (3) view and traverse their list of connections and those made by others within the system'.

the wearable AR devices: first, the wearable AR devices are ‘always-on’ devices, which means that everyone who wears such a device can film or photograph continuously, and secondly, these devices do that surreptitiously without anyone taking notice of the action and without anyone being able to consent or not.⁶¹

What follows now is a more detailed description of the actions and the types of privacy that these actions can potentially infringe.

3.4.1 Capturing photos and videos

Without the use of wearable AR devices, if a person wants to take a photo of another person with a mobile telephone or a camera in public, the user of the device has to take the device out of the pocket or the bag, hold the device, focus towards the subject and then take the photo. These are obvious movements, making it visible to people around that somebody is taking photos at that time. With a wearable AR device, however, this action becomes totally seamless and surreptitious; the device is on the head of the user and is pointing in the exact same direction as the head of the user, thus making the procedure of taking a photo or filming imperceptible to the third person who becomes the subject of the photo or the film. Google has already introduced the ‘wink’, which means that the user can take photos by just winking, something which makes the action of photographing even more difficult to be noticed.⁶²

The ability of wearable AR devices to capture images and videos seamlessly and surreptitiously could be compared to some existing forms of stealthy surveillance such as the Unmanned Aircraft Systems (UAS). The privacy issues arising from the UAS devices are that they create a state of surveillance, in the same way as Closed Circuit Television (CCTV), but they do that covertly, without noise, invisibly and therefore without individuals knowing that they are being monitored.⁶³ In the same way, wearable AR devices are able to monitor third persons without their knowing that they are being monitored: a wink by the user is enough to start the monitoring of a third person. In addition, when wearable AR devices become a mainstream technology, no one will really notice them, since people will be so used to the device. There is, however, one difference: UAS devices are used by the government or some private entities, such as magazines or private security, for surveillance⁶⁴; wearable AR devices, even though they can be used for surveillance by the government or companies, they can also be used by any individual. That, as Steve Mann argued, will create a society not just of surveillance (*sur*: above, and *veiller*: to watch), where devices, such as CCTVs and UASs, are monitoring, from above, individuals for the security

61 See also Henry B Strub and others, ‘Privacy. Wearable Computers, and Recording Technology’ 150 Digest of Papers (Second International Symposium on Wearable Computers, 1998), 19-20 October 1998, Pittsburgh, PA, USA where the impact on privacy is discussed by members of the wearable computer community.

62 ‘Google Glass Takes Photos by Winking’ BBC (18 December 2013), London, United Kingdom <<http://www.bbc.co.uk/news/technology-25426052>> accessed 27 December 2013.

63 Finn (n 54) 10.

64 See ‘Unmanned Aerial Vehicle’ (2013), Wikipedia, the free encyclopedia <http://en.wikipedia.org/w/index.php?title=Unmanned_aerial_vehicle&oldid=587880088> accessed 27 December 2013. It should be mentioned here, however, that the number of drones operated by civilians has increased lately and many issues discussed in this article regarding the privacy implications by the use of wearable AR devices are similar.

of the state or for companies, but also a society of *sousveillance* (*sous*: below), namely a society where everyone can monitor an activity by being part of the activity, or, expressed differently, where everyone can monitor everyone else.⁶⁵

From the seven categories of types of privacy described above, the ones that can be impacted by the capturing of videos and photos are the privacy of behaviour and action, privacy of data and image, privacy of location and space, privacy of association and privacy of thought and feelings.

As far as the privacy of behaviour and action of the third person is concerned, this technology has made it possible for everyone to be monitored and recorded unaware. If a situation like this, where everyone is 'sousveilled' continuously were the rule, then individuals would start to adjust their behaviour, since they would assume that someone was always monitoring them and that their every move could possibly be made public by someone uploading a recording to the Internet. The situation reminds us of Bentham's Panopticon, where the prisoners knew that they all were continuously the subjects of 'virtual observation without ever being able to confirm its actuality'⁶⁶ in the same way that individuals would be subjects of monitoring by other individuals without ever being able to confirm it. This constant awareness of being possibly monitored could lead to anticipatory conformity, where individuals would behave differently from how they would otherwise behave because they believe that someone is monitoring and recording them at all times.⁶⁷

In the same spirit, this constant state of 'sousveillance' can impact the privacy of data and image. The image of an individual, which is captured, digitized, stored and processed by a wearable AR device, constitutes part of this person's personal data. As Wright and Raab stated, individuals should always be able to know what data are made available and to which other individuals or organizations.⁶⁸ Since wearable AR devices can capture surreptitiously the image of a third person, this person cannot 'exercise a substantial degree of control over that data and its use'.⁶⁹

Furthermore, wearable AR devices can also connect a person to a specific time and a specific space, since they can add a time and location stamp on the image or video that is captured. This could impact the privacy of location and space as this person can no longer 'be present in a location or space without being tracked or monitored or without anyone knowing where he or she is'.⁷⁰ Additionally, information about private spaces, such as a home, can be revealed if the user of a wearable AR device is near or in this private space.

The use of wearable AR devices may also impact the privacy of association since the videos or photos can capture a third person interacting with other individuals. Affiliations and relations this person would like to retain private can be discovered

65 Mann (n 8).

66 Paulo Vaz and Fernando Bruno, 'Types of Self-Surveillance: From Abnormality to Individuals' at Risk' (2003) 1(3) Surveillance & Society 275.

67 Finn (n 54) 11.

68 Wright and Raab, 'Privacy Principles, Risks and Harms' (2014) 28(3) International Review of Law, Computers & Technology 277 <<http://www.tandfonline.com/doi/pdf/10.1080/13600869.2014.913874>> accessed 12 January 2015.

69 Nissenbaum (n 49) 82.

70 *ibid.*

and made public. The phenomenon of anticipatory conformity may present itself in this situation too since people may want keep some affiliations and relations private. As Fin Wright and Friedewald stated, everyone must have the 'right to associate with whomever they wish'⁷¹ to pertain a democratic society.

Lastly, the capturing of images and videos may also reveal the thoughts and feelings of a person. The way people walk, for example, can reveal their psychological condition, such as being happy or depressed.⁷² Likewise, as it will be noted later on this article, the face can also reveal the emotions of a person. These emotions can be then used for profiling and even for manipulative decision-making purposes.⁷³

3.4.2 *Uploading the data on the web*

In the case that a user of a wearable AR device uploads on a SNS or a website the image or video of a third person, then all his/her information included in this image or video will potentially become accessible to everyone in the world. The images and videos could then be further processed to extract even more information about him/her.

The types of privacy that were mentioned above would thus be also impacted in this scenario, since anyone with access to the SNS or website could process the data in the same way as the user of the wearable AR device: data such as the image of this person, the time and location of his/her actions, the people he/she interacted with and his/her psychological condition, will be available to potentially everyone with Internet access. Furthermore, anyone will be able to further process these data and with help of biometrics extract even more data.

Therefore, it can be stated that the uploading of such data on the web constitutes a multiplied effect on the impact onto the types of privacy that were mentioned in the chapter above since the data become available to a vast amount of people.

3.4.3 *Biometrics*

The third action that is connected to the use of wearable AR devices is face recognition and biometrics, in general. By way of illustration: somebody walks in the street with a wearable AR device and can in real time identify the third persons that he/she is looking at by using face recognition software, or he/she can take a photo of a person and later proceed to identification of that person by the same technique.

Face recognition, as well as most of the other biometrics such as body recognition, consists of six sub-processes: (i) image acquisition, namely the process of capturing an image of an individual, (ii) face detection, (iii) normalization, (iv) feature extraction, (v) enrolment, namely if it is the first time that the subject of the recognition is processed the image will be stored for feature comparison and (vi)

71 Finn (n 54) 5.

72 Michalak and others, 'How We Walk Affects What We Remember: Gait Modifications Through Biofeedback Change Negative Affective Memory Bias' (2015) 46 *Journal of Behavior Therapy and Experimental Psychiatry* 121 <<http://www.sciencedirect.com/science/article/pii/S0005791614000809>> accessed 15 January 2015.

73 Finn (n 54) 17.

comparison with other images in the databases.⁷⁴ In the last sub-process, categorization of the features of the subject is taking place to classify them and make the processing faster. That means that to proceed to face recognition, even if this is happening in real time, an image will be captured and stored.

Face recognition and biometrics, in general, have raised many privacy issues. As Venier and Mordini argue, the new biometrics could take place remotely, and covertly, extracting much sensitive information about an individual.⁷⁵ Even though the face recognition is still considered an emerging technology, the issues that it raises are already here. Although Google has stated that for now they will not allow face recognition applications for the Google Glass, the situation will change when other companies will produce wearable AR devices and when these devices will begin to be 'rooted' (when the users start giving themselves privileged control of the device, thus overcoming the limitations put by the producers of the operating systems and devices⁷⁶). In short, this is a problem that will arise sooner or later and if we think about the fact that 250 million photos are uploaded everyday on Facebook alone⁷⁷ we can understand that the 'pools', the databases, which the face recognition applications will use to identify persons, are already available. It is worth mentioning here that the pools have been created to be monetized at some point, since the business model of SNSs, such as Facebook and Google+, is to monetize the data of their users.⁷⁸

Additionally, experiments have shown that because of the huge amount of photographs that exist on the Internet, face recognition applications can already identify quite precisely the person whose face is captured by a photo.⁷⁹ Specifically, an experiment led by Alessandro Acquisti at Carnegie Mellon University used an off-the-shelf software to find information that was available on the Internet. The design of the experiment was that students would let the researchers photograph them and, by using the software, try to find the identity of the students. The experiment was quite

74 Working Party (WP) 129, 00727/12/EN, Opinion 2/2012 on facial recognition in online and mobile services, 2.

75 Silvia Venier and Emilio Mordini, 'Second-Generation Biometrics' in Rachel Finn and David Wright (eds), *Privacy, Data Protection and Ethical Issues in New and Emerging Technologies: Five Case Studies* (PRESCIENT consortium, 25 November 2011).

76 'Android Rooting' (2013), Wikipedia, the free encyclopedia <http://en.wikipedia.org/w/index.php?title=Android_rooting&oldid=587671795> accessed 27 December 2013.

77 Ben Parr, 'Facebook by the Numbers [INFOGRAPHIC]' (2011), Mashable <<http://mashable.com/2011/10/21/facebook-infographic/>> accessed 12 December 2013.

78 Amy A Hinkler, 'Privacy Issues in the Age of Social Media M&A' (2013) <http://works.bepress.com/cgi/viewcontent.cgi?article=1000&context=amy_hinkler> accessed 16 January 2014.

79 In the article by Mark Williams on May 30 and 2007, 'Better Face-Recognition Software' <<http://www.technologyreview.com/news/407976/better-face-recognition-software/>> accessed 14 December 2013. we can see that even the face recognition still has many drawbacks the potentials are getting better and better considering that in 2007, when the article was written, 'the match -up of face recognition algorithms showed that machine recognition of human individuals has improved ... hundredfold since 1995', namely in 12 years. Furthermore, software for face recognition is already used by some authorities, such as the police in San Diego (eg Ali Winston, 'Facial Recognition, Once a Battlefield Tool, Lands in San Diego County' (07 November 2013), The Center for Investigative Reporting <http://cironline.org/reports/facial-recognition-once-battlefield-tool-lands-san-diego-county-5502?utm_source=CIR&utm_medium=social_media&utm_campaign=tumblr> accessed 5 December 2013).

successful even though as Acquisti said, '[t]he capabilities of automated face recognition *today* are still limited - but keep improving.'⁸⁰

The first type of privacy that is potentially impacted by the use of this technology is the privacy of the person because of the measurement and digitization of the body and face. Furthermore, the personal data and image privacy is also impacted because individuals are identified and categorized, since they are linked to an existing profile, such as a SNS profile, which can have further information of that person varying from his/her favourite books to his/her political beliefs.⁸¹ The threat for both these types of privacy comes from the fact that this technology makes possible the 'systematic collection of information that could be used for classification purposes'⁸²; and this can take place without the knowledge or consent of the individual. In addition, they can reveal a great deal of sensitive personal information such as ethnicity, age and gender. A full-scale profile could be made for every individual.

Privacy of behaviour and actions, as well as of thought and feelings can also be infringed since biometrics has the ability to extract psychological information about the person identified.⁸³ Psychological and behavioural profiling can thus be done to be used for behavioural prediction.⁸⁴

The privacy of location and space can also be impacted, since when a person becomes identifiable it can be connected directly to a specific time and place because of the time and place stamps. After a person is identified he/she could be easily tracked or even stalked.⁸⁵

Now that we have described if and how the use of wearable AR devices can have an impact on the seven types of privacy, the existing legislation for the protection of privacy will be examined.

4. THE LAW

Privacy, as already stated, is connected to social structures, political and economic ideologies and national traditions.⁸⁶ Therefore, in different countries the legal concept of privacy has been materialized in different ways. Most noticeable is the difference between the concepts of privacy as implemented in the common law right to

80 'Face Recognition Study – FAQ' 2011 <<http://www.heinz.cmu.edu/~acquisti/face-recognition-study-FAQ/>> accessed 11 December 2013.

81 For example, 'Facebook' (2013), Wikipedia, the free encyclopedia <<http://en.wikipedia.org/w/index.php?title=Facebook&oldid=587882105>> accessed 27 December 2013 stating that on Facebook the user is inserting list of personal interests and information.

82 Finn (n 54) 16.

83 See Cotton Delo, 'Microsoft Files Patent to Serve Ads Based on Mood, Body Language' (12 June 2012), Ad Age <<http://adage.com/article/digital/microsoft-files-patent-ad-serving-tech-senses-mood/235336/>> accessed 27 December 2013, where it is stated that Microsoft has already patented an advertising engine based on facial expressions.

84 Finn(n 54).

85 See Woodrow Hartzog, 'I See You: The Databases That Facial-Recognition Apps Need To Survive' (23 January 2014), Center of Internet and Society, Stanford Law School <<http://cyberlaw.stanford.edu/publications/i-see-you-databases-facial-recognition-apps-need-survive>> accessed 20 January 2015.

86 See Peter Blume, 'Data Protection and Privacy – Basic Concepts in a Changing World' in *Scandinavian Studies in Law Volume 56, ICT Legal Issues* (2010) Jure Law Books, Stockholm, Sweden.

privacy in the USA, and the data privacy as implemented in the data protection legislations across the EU countries.⁸⁷ Since these are the two territories/markets where Google Glass and other wearable AR devices are expected to target initially, the article will now focus on what laws exist in both the USA and EU to protect the privacy of third persons from the user of wearable AR technologies.

4.1 The US approach

Privacy protection, in general, under the US law could be described as a complex system that consists of constitutional protection, federal and state statutory protection as well as tort protection.⁸⁸

Since the focus of the article is the protection of privacy of third persons from the users of wearable AR devices and, to some extent, other private entities, no extensive reference will be made to the protection of privacy by governmental intrusions, even though it is possible that governmental agencies will use these devices to gather data. I would like to point out, however, that the US Constitution—as well as a number of state and federal laws⁸⁹—does provide protection against certain governmental intrusions; nonetheless, it does not declare a right to privacy against private intrusions.⁹⁰

In lack of such a right, protection can be sought primarily in tort law. Since the declaration of the ‘right to be let alone’ by Warren and Brandeis, most states recognized a common law right of privacy through the state privacy common law torts. The torts that are most commonly recognized and are stated in the Restatement (Second) of Torts, are the ones that William Prosser, reporter for the Restatement, introduced in 1960 when he stated that privacy as a legal concept has come to constitute four distinct torts: (i) intrusion upon seclusion, (ii) public disclosure of private facts, (iii) false light and (iv) appropriation.⁹¹

The question that rises here is if these torts can really protect the privacy of third persons from other individuals/users of wearable AR devices.

As far as the capturing of video and photo is concerned there are two distinctive cases. One is when the capturing takes place in a private area and the other when it takes place in a public space.

For the privacy intrusion that takes place in a private space the torts seem to provide adequate protection since if an intrusion takes place, the third person could seek recovery for the harm either because his/her solitude was intruded or because private facts were disclosed. It should be noted, however, that even in such cases the outcome can be quite unpredictable. As an example in *Plaxico v Michael*, the Mississippi Supreme Court stated that even in the case of intrusion upon the solitude or

87 See Lee A Bygrave, ‘Privacy and Data Protection in an International Perspective’ in *Scandinavian Studies in Law Volume 56, ICT Legal Issues* (2010) 166, Jure Law Books, Stockholm, Sweden about the term ‘data privacy’.

88 Jon L Mills, ‘Privacy: The Lost Right’ (2008), Oxford Scholarship online <<http://www.oxfordscholarship.com/view/10.1093/acprof:oso/9780195367355.001.0001/acprof-9780195367355-chapter-4>> accessed 9 January 2015.

89 See 5 USC s 552a (2000) and Pub L 107–347 (E-Government Act 2002).

90 Amy A Hinkler, ‘Privacy Issues in the Age of Social Media M&A’ (2013) 3 <http://works.bepress.com/cgi/viewcontent.cgi?article=1000&context=amy_hinkler> accessed 16 January 2014.

91 William L Prosser, ‘Privacy’ (1960) 48(3) California Law Review, 389.

seclusion of a person, the act must be ‘a substantial interference’ and ‘highly offensive to the ordinary, reasonable man’. Therefore, it found that the act of Mr Michael, namely his taking a photo through the window of a cabin where Ms Plaxico, who had an affair with Mr Michael’s ex-wife, was half naked, was not intruding her privacy. The reason was that most reasonable people, according to the court, would have found Mr Michael’s action was not substantially invasive and it was justified since he did it to gain custody of his child.⁹²

Regarding the protection of privacy in public spaces it could be stated that in the USA, the tradition of privacy protection seems to be originating from the old dictum by Sir Edward Coke that ‘my home is my castle’.⁹³ That means that privacy is connected only to the private space of an individual and there is no expectation of privacy in the public sphere. According to Warren and Brandeis, the right to privacy is the right to be let alone in ‘private life’,⁹⁴ in a secluded place such as the home of a person. Furthermore, as Prosser stated, there cannot be expected to be any right to privacy ‘[o]n the public street or in any other public space.’⁹⁵ He continues by saying that ‘anything visible in a public place may be recorded and given circulation by means of a photograph, to the same extent as by a written description, since this amounts to nothing more than giving publicity to what is already public and what anyone present would be free to see’.⁹⁶

Furthermore, another doctrine that is widely used in the USA, the ‘reasonable expectation of privacy’ doctrine, as stated in the Katz case,⁹⁷ refers only to governmental intrusion in public spaces where a reasonable person would expect the same privacy as in his/her home, such as public restrooms.⁹⁸

Therefore, it seems rather doubtful that the privacy of an individual will be protected under tort law in the case of simple—non ‘substantially interfering’⁹⁹—video recording or photographing in a public space.

Regarding the uploading/publishing of the captured videos and photos the exact same problem arises, since publishing something that has taken place in a public space does not constitute privacy intrusion, according to the same reasoning that was described above. It is worth mentioning here that in two cases of videotaping and publishing of a video on TV—a situation that is quite similar to uploading a video or photo on the Internet after recording it with a wearable AR device—US courts found that the plaintiffs could not be protected under public disclosure of private facts or intrusion upon seclusion.¹⁰⁰ Similarly, the Washingtonienne case raised

92 *Plaxico v Michael* (Miss 1999) 735 So 2d 1036, 1038.

93 See the Assembly No 159 State of New Jersey, 213th Legislature, The New Jersey Self Defense Law, where it is stated that ‘[t]he “castle Doctrine” is a long-standing American legal concept . . . that provides that one’s abode is a special area in which one enjoys certain protections and immunities.’

94 Warren and Brandeis (n 43) 215.

95 Prosser (n 91) 391.

96 *ibid* 394.

97 *Katz v United States* (1967) 389 US 347.

98 See also Robert McArthur, ‘Reasonable Expectations of Privacy’ (2001) 3 Ethics and Information Technology 123.

99 *Plaxico* (n 92).

100 *Gritzke v MRA Holdings, LLC*, No 4:01CV495-RH, 2002 WL 32107540 (ND Fla, 15 March 2002), where the court found that the use of an embarrassing picture of a young woman on a video without her

the issue of unsuitability of the tort of public disclosure of private facts in cases where two people or more share a moment.¹⁰¹

Since, as seen above, protection under tort law in the case when an individual wants to protect his/her privacy from users of wearable AR devices looks quite troublesome and doubtful, the possibility to seek protection under federal and state statutes and regulations will be examined. These statutes and regulations were created to close some of the gaps in tort law.¹⁰² It should be noted, however, that there are 'more than forty federal statutes, multiple regulations, several state constitutions and hundreds of state statutes tackling specific types of intrusions',¹⁰³ something that adds to the complexity of the US system of privacy protection. The reason is that in the USA privacy is approached on an 'issue-by-issue'¹⁰⁴ base with sector-specific rules.

Mill counts 10 different categories of federal and state statutes related to privacy.¹⁰⁵ The one that is most relevant in our scenario is the category of statutes protecting against private intrusions. This category includes the Video Voyeurism Prevention Act 2004¹⁰⁶ and stalking statutes,¹⁰⁷ as well as the cyberstalking statute.¹⁰⁸ The problem, however, that arises in this situation is that both federal and state statutes ask for 'repeatedly', 'wilful' and 'malicious'¹⁰⁹ infringement of one's freedom. Furthermore, they prohibit video recording or photographing without consent only in situations where a person has a reasonable expectation of privacy, creating thus the same problem as in the case of the torts that was described above.¹¹⁰ Therefore, in a scenario of a one-time video recording that is uploaded on an SNS these statutes are hard to apply.

It should be mentioned here that in addition to the statutory protection, there is also a regulatory protection through agencies such as the Federal Trade Commission (FTC).

The FTC is responsible for monitoring private companies against unfair and deceptive acts and practices.¹¹¹ This means that the FTC is approaching privacy as a consumer's right. It does not, however, have any court power, and therefore its role is limited to 'encourage the industry to self-regulate' and to litigate 'on behalf of the

consent for the release of that video had no remedy under public disclosure of private facts or intrusion upon seclusion but only under the right to publicity because her image was on the cover of the videotape and on the advertisements. See also *Lane v MRA Holdings, LLC*, 242 F Supp 2d 1205 (MD Fla 2002).

101 *Steinbuch v Cutler*, No 05CV00970, 2005 WL 1467405 (DDC 18 May 2005). See also the discussion about this case by Mills (n 88) 261.

102 Mills, *ibid* 130.

103 *ibid*.

104 *ibid*.

105 *ibid* 135.

106 Video Voyeurism Prevention Act of 2004, 18 USC s 1801.

107 For a state-by-state database of state stalking statutes, see Stalking Resource Center <<https://www.victimsofcrime.org/our-programs/stalking-resource-center/stalking-laws>> accessed 20 January 2015. See also the federal statute 18 USC s 2261A(1) (2000).

108 18 USC s 2261A(2) (2000).

109 Mills (n 102) 154.

110 See also *State v Glas*, 147 Wash 2d 410, 414 (2002), where a court held that a state voyeurism statute only covered locations where a victim would have an expectation of privacy. Therefore, the statute failed to prohibit 'up-skirt' photography in a public location.

111 See 15 USCA s 57a (West).

people'.¹¹² Therefore, an individual cannot bring a private action to the FTC to compensate for injuries that he/she suffered because of privacy intrusion. Furthermore, the FTC is responsible only for intrusions by companies and not by individuals.

Lastly, as far as biometrics is concerned, in the USA there is no specific legislative framework that regulates this technology as a whole except some laws, such as the Illinois Biometric Information Privacy Act of 2008 that addresses the collection, use, safeguarding, handling, storage retention and destruction of biometric identifiers and applies to private entities.¹¹³

Otherwise, it is the FTC that supervises how biometrics is implemented in the market. Therefore, it has issued some guidelines to 'persons, partnerships, or corporations'¹¹⁴ who process biometrical data.¹¹⁵ As FTC states, 'companies should not use facial recognition to identify anonymous images of a consumer to someone who could not otherwise identify him or her, without obtaining the consumer's affirmative express consent'.¹¹⁶ Therefore, only individuals who have agreed to participate in face recognition should be identified. Furthermore, 'even if a company does not itself intend to implement facial recognition technologies, it should consider putting protections in place that would prevent unauthorised scraping of the publicly available images it stores in its online database'.¹¹⁷ The problem that arises, though, is that the guidelines are not obligatory. Only '[if] a company uses facial recognition technologies in a manner that is unfair . . . or that constitutes a deceptive act or practice, the Commission can bring an enforcement action under Section 5'.¹¹⁸ According, however, to 15 USC section 45(a) an unfair act is only an act that can cause a 'substantial' injury to consumers. What constitutes substantial injury is examined by case, thus creating an unclear framework of protection. According to FTC's policy statement on unfairness, 'substantial' may be an injury that involves 'monetary harm' and 'unwarranted health and safety risks' but normally not 'emotional impact and other more subjective types of harm'.¹¹⁹ The last type of injury is, nonetheless, the common type of injury in our scenario. Therefore, as Thomas claimed '[t]he boundaries of unfairness, as applied to Internet privacy violations, remain an open question'.¹²⁰

112 Amy A Hinkler, 'Privacy Issues in the Age of Social Media M&A' (2013) 24 <http://works.bepress.com/cgi/viewcontent.cgi?article=1000&context=amy_hinkler> accessed 16 January 2014.

113 Biometric Information Privacy Act, 740 ILCS 14. See also Kathy Gurchiek, 'Facial Recognition Technology Raises Privacy Issues' (14 September 2012), Society for Human Resource Management <<http://www.shrm.org/hrdisciplines/technology/articles/pages/facial-recognition-privacy.aspx>> accessed 12 December 2013.

114 15 USC s 45(a)(1) (2000).

115 Federal Trade Commission, 'Facial Recognition: Best Practices for Common Uses of Facial Recognition Technologies' (2012) <<http://www.ftc.gov/sites/default/files/documents/reports/facing-facts-best-practices-common-uses-facial-recognition-technologies/121022facialtechrpt.pdf>> accessed 15 December 2013.

116 *ibid* iii.

117 *ibid* 17.

118 *ibid* 2.

119 'FTC Policy Statement of Unfairness' (1980) <<http://www.ftc.gov/public-statements/1980/12/ftc-policy-statement-unfairness>> accessed 15 January 2015.

120 Leary B. Thomas, 'Unfairness and the Internet', Federal Trade Commission (2000), <<http://www.ftc.gov/public-statements/2000/04/unfairness-and-internet>> accessed 12 January 2015.

To sum up: in the absence of a constitutional right to privacy against private intrusions an individual can seek protection under tort law or some federal and state statutes. In cases where the intrusion took place in a private space, tort law is quite clear and most of the times no problems seem to arise. In cases where the intrusion happened in public space, however, tort law seems to be inadequate to provide protection to the individual. And even if there are some federal and state statutes that could be used by an individual in some cases, such as if the user of a wearable AR device is stalking or harassing a third person, the rule under US privacy law remains that things that happen in public are not worthy of protection with regards to privacy.

4.2 The EU protection

On the contrary, the protection of privacy is approached quite differently under EU law. On one side, there is a general right to respect private life of individuals dictated by the Charter of Fundamental Rights Article 7,¹²¹ which became part of the Treaty on European Union (TEU) and follows the same line as the European Convention of Human Rights (ECHR) Article 8, which all European countries have ratified. That means that Member States do not only have the obligation to not intrude in the private life of their citizens, but also to create the proper conditions so that their citizens can enjoy their rights in general. On the other side, modern forms of communication lead the EU to specify this right to privacy regarding the processing of individuals' personal data under the Data Protection Directive.¹²²

The Directive, which has become the main instrument for privacy protection in Europe, has as its main concept that in the modern era every process of personal data, namely any operation such as 'collection, recording, organization ... or destruction'¹²³ of 'any information relating to an identified or identifiable person',¹²⁴ can lead to privacy violation. As stated in the recital 26 of the Directive, all data that makes a person identifiable can be considered personal data. Therefore, all data that can be extracted by a photo or by a video are considered personal and thus protection is acquired by the Directive.

There are, however, some criteria that make the data processing legitimate according to Article 7. From these, the one that is of importance for the scenario of this article, namely the protection of privacy of third persons from the users of wearable AR devices and, to some extent, other private entities, is the first one, namely the consent of this person.¹²⁵

According to the Directive consent of the person/data subject, is necessary to be obtained before his/her data is gathered, otherwise the gathering and processing of

121 Charter of Fundamental Rights of the EU (2010/C 83/02).

122 Under art 8 of the Charter, the right to the protection of personal data has become a fundamental right.

123 Council Directive 95/46/EC of the 24 October 1995 on the protection of individuals with regard to the processing of personal data and on the free movement of such data [1995] OJ L281/31, art 2(b).

124 *ibid*, art 2(a).

125 The scenario of this article is focusing on the protection of privacy of the third person, namely not the user of the wearable AR device or other entities that may process the data, and therefore the exceptions provided concerning contractual relationships (under art 7(b) of the Directive), governmental actions (under art 7(c) and (d) of the Directive) and data that after manifestation became public (under art 7(f) of the Directive) are of no interest.

the data is illegitimate.¹²⁶ This consent must, furthermore, be explicit for some special categories of data, including data revealing political opinions, religious or philosophical beliefs, union membership and health or sex life.¹²⁷

In our scenario now, for a third person to seek protection under the Data Protection Directive, the most important task is to find out who is the data controller, namely who gathers and processes the data.

It should first be noted here that in the case that the user of a wearable AR device captures a video or photo of a third person, and the processing of the data takes place ‘in the course of a purely personal or household activity’ the Directive does not apply¹²⁸; national laws, however, created in accordance to the Charter and the ECHR are applicable.¹²⁹ It should also be reminded here that the European Court of Human Rights (ECtHR) has many times found that photographing without one’s consent is a violation of Article 8 ECHR, namely the right to respect for one’s private life as it is specialized into the right to protection to one’s image. According to the ECtHR, domestic courts have to take into consideration if consent is given. Furthermore, there is no need to prove that such an action created damage but the mere fact that no consent was given is enough.¹³⁰

If the user of the wearable AR device uploads the video or photo to his/her own website or an SNS platform, he/she becomes the controller if he/she ‘acts on behalf of a company or association, or uses the SNS mainly as a platform to advance commercial, political or charitable goals’ as well as ‘when access of profile information extends beyond self-selected contacts’.¹³¹ The access is considered as extended when the data become widely public, either to all SNS users or to all web users, through indexing of the data by search engines and even when the data are available to self-selected contacts alone but the number of these contacts is high. Furthermore, after the uploading to an SNS, the SNS provider also becomes a controller according to the Directive if it collects and processes the uploaded data, namely all data that can be extracted by the photo or video, such as the geo-location and time stamps and biometrics.¹³²

The data controller—the company, the user or both¹³³—is now liable if the data is collected or processed without the third person’s/data subject’s consent. If these

126 See n 123, art 7(a).

127 *ibid*, art 8(2a).

128 See Household Exception of the Directive, art 3(2).

129 For example, in Germany, there is the constitutional right to personality (Persönlichkeitsrecht) and the right to the persons image (das Recht am eigenen Bild); in Greece the constitutional right to personality (το δικαίωμα στη προσωπικότητα); in France the right is protected in the Civil Code under art 9, which does not allow photographing or videotaping without consent.

130 For a list of cases related to art 8 ECHR, see ECHR, ‘Factsheet – Right to the Protection of one’s Image’ (2014) <http://www.echr.coe.int/Documents/FS_Own_image_ENG.pdf> accessed 10 January 2015.

131 WP 163, 01189/09/EN, Opinion 5/2009 on online social networking, 6.

132 *ibid* 6 and 12.

133 It must be stated here that the word ‘both’ that is used here, does not have the meaning of the Directive’s ‘jointly’ (see art 2(d) of the Directive). It just states the fact that it is possible to have at the same time two different controllers. See also the WP 169, Opinion 1/2010 on the concepts of ‘controller’ and ‘processor’ Part III(1)(d) example No 7, where it is mentioned that there can be different data controllers on the same time.

data are sensitive, as the ones extracted after the use of biometrics, the consent has to be explicit.

The Directive, in combination with the Charter and the ECHR, creates a legal framework of protection of privacy, which is a lot more complete than the one provided in the USA. The reason is that the Directive does not treat privacy in public space differently than privacy in private space and it does not differentiate online privacy to consumers' and non-consumers' privacy, as is the case in the USA. Furthermore, the Directive gives a two-folded protection to the data subjects: an individual can lodge a complaint either with the national supervisory authorities (DPAs), which are the main actors aiding individuals and enforcing the data protection legislation and have powers that include monetary and pecuniary sanctions, or through judicial procedures, to seek compensation in the form of damages.

Thus, it could be stated that the European approach creates just one framework, whereas the US approach creates many different frameworks depending on whether the privacy needs to be protected from the government from companies or from individuals.¹³⁴

Even though the Directive could prove to be a good tool for privacy protection it also has a big drawback. It has created a fragmented legal framework throughout the EU countries since it is transposed differently in every EU country. This has led to a general confusion both for individuals as well as for companies that have to bear a significant cost for compliance with all the fragmented legislations. Furthermore, the sanctions and remedies mentioned above vary from country to country.¹³⁵

These problems, though, are intended to be addressed when a regulation, which is already proposed, will be enacted throughout all EU countries, homogenizing privacy legislation in the EU.¹³⁶

To conclude, the EU approach on privacy protection focuses on the protection of all the data of a person without distinguishing between private and public sphere. All data could possibly be used for privacy violation and thus they are worthy of protection without the need to prove damage. Therefore, for the user of a wearable AR device to take photo or video of a third person and upload it on the Internet, consent has to be given by the third person. Furthermore, for face recognition to take place, the third person has first to give his/her explicit consent in order for his/her data to be processed so that he/she can be identified.

5. RECOMMENDED SOLUTIONS

In this final section, some solutions will be proposed to create an effective framework for protection of privacy regarding the use of wearable AR devices, based on law, social norms, company policies and technology. First, a recommendation will be made

134 Protection is provided, as already stated in the previous section of this article, respectively, by the Constitution, the FTC and federal and state laws.

135 For a more detailed description of the remedies and the sanctions throughout the EU countries, see EU Agency for Fundamental Rights, 'Access to Data Protection Remedies in EU Member States' (2013), Publications Office of the European Union <http://fra.europa.eu/sites/default/files/fra-2014-access-data-protection-remedies_en.pdf> accessed 10 January 2015.

136 See also Alexander B Blumrosen, 'The Proposed EU Data Protection Regulation' (2012) 41(4) *International Law News* 20.

with regard to the US approach on privacy, which, as mentioned above, has proven obsolete and fragmented. Next, two technical solutions will be recommended. Both are influenced by the social translucent systems, described by T Ericsson and WA Kellogg, according to which in order for a digital system, in our situation a technical solution, to be efficient three characteristics have to be met: visibility, awareness and accountability.¹³⁷

5.1 Redefining privacy

As mentioned above, the biggest problem with the US approach is that privacy protection in public spaces is absent and that there is no comprehensive national privacy legislation that could be used by an individual against privacy intrusions by non-governmental actions. Therefore, a privacy right to public spaces has to be recognized. The old mentality, which created a tradition lasting almost 120 years, stating that whatever happens in the public sphere is not worth to be protected, cannot be continued in the era of ubiquitous computing¹³⁸ where every move and every data of a person is traceable and where everyone can continuously and surreptitiously take photos and videos of other people, upload them on the Internet without their consent, thus making the photos, the videos and the data that can be extracted out of them, available to the whole world.

As a first step, privacy, as it has been interpreted until now by tort law, has to be redefined to include also privacy in the public sphere. As McClurg stated, '[p]rivacy is a matter of degree.'¹³⁹ That means that the existing approach to privacy as an all—if in a private space—or nothing—if in a public space—approach is not valid; when a person accepts to give up some of his/her privacy by being in a public space, that does not mean that this person accepts to forfeit all his/her privacy.¹⁴⁰ Arguments used in the past, for example, that taking a photograph or video of a person in a public space equals to taking a record of a moment that either way anyone could see,¹⁴¹ are outdated: an instantaneous mental picture is not the same as a permanent digital picture. The latter can be further studied and processed so that more information can be extracted, for example, the mood of the subjects of the photograph. In addition, the audience of a photo or video uploaded on the Internet is not only much larger than the audience that witnessed the moment that was captured by a photo or video, but it is also a different audience, consisting of people from every place of the world.¹⁴² There is both a qualitative and quantitative difference, and this is the reason why the surreptitious capturing of a photo or video by a wearable AR device should be treated differently.

In the same spirit, Nissenbaum claims that the dichotomy of public/private does not function anymore and, that in an era where all kind of information can be

137 Thomas Erickson and Wendy A Kellogg, 'Social Translucence: An Approach to Designing Systems That Support Social Processes' (2000) 7(1) ACM transactions on computer-human interaction (TOCHI).

138 See n 46.

139 Andrew Jay McClurg, 'Bringing Privacy Law out of the Closet: A Tort Theory of Liability for Intrusions in Public Places' (1994) 73 North Carolina Law Review Association 1041.

140 *ibid*; See also W Ward Fearnside and William B Holther, *Fallacy: The Counterfeit of Argument* (1959).

141 Posser (n 91) 392.

142 McClurg (n 139) 1043.

aggregated and analysed in different contexts, even public information can lead to privacy violation since the information of one context, namely information related to the public sphere which is not *per se* privacy intrusive, can be used in another context, and therefore violate privacy.¹⁴³ Or as expressed differently by Calvert and Brown: some matters are private even in public space.¹⁴⁴

As a second step, a federal legislation is needed.¹⁴⁵ Instead of seeking privacy protection under tort law, trying to prove monetarily the amount of damage a person suffered, which is extremely difficult in privacy violation cases where the damage is not tangible, violation of privacy should be enough for somebody to seek protection.¹⁴⁶

A problem that has already been stressed in this article is that the legal conceptualization of privacy is a very difficult, if not impossible, task. It is connected to many different factors and is more of subjective nature. On the contrary, data protection has specific and objective principles that can provide for 'normative guidance' to courts and authorities¹⁴⁷; and even if it is narrower than the concept of privacy,¹⁴⁸ it is closely connected to issues arising by wearable AR devices, since most concerns, as described above, are connected to information/data of a third person that are gathered and processed without the person's consent.

Therefore, it is proposed here that the US legislators approach privacy through a data protection prism as the EU has already done. This approach should be considered the most pragmatic approach: all data are treated as private and worthy of protection no matter the context, forum, actor or place, since every data could lead to violation of privacy if combined with other data or used in a different context.¹⁴⁹ By the same token, a data protection approach by the USA would contribute to the harmonization of privacy protection rules on a global level since the different approaches in these two main technology markets, USA and EU, create problems both for the companies that have to implement multiple privacy policies for every market as well as for enforcement authorities.¹⁵⁰

5.2 A priori protection

One problem with the protection of privacy of a third person from the use of wearable AR devices is that the protection provided through legislation is provided mostly *a posteriori*, namely after a violation has taken place, that is, after the user of the

143 Nissenbaum (n 49).

144 Clay Calvert and Justin Brown, 'Video Voyeurism, Privacy, and the Internet: Exposing Peeping Toms in Cyberspace' (2000) 18 Cardozo Arts & Entertainment Law Journal 469.

145 Amy A Hinkler, 'Privacy Issues in the Age of Social Media M&A' (2013) 41 <http://works.bepress.com/cgi/viewcontent.cgi?article=1000&context=amy_hinkler> accessed 16 January 2014; See also Mills (n 102) 273–75.

146 Amy A Hinkler, 'Privacy Issues in the Age of Social Media M&A' (2013) 42 <http://works.bepress.com/cgi/viewcontent.cgi?article=1000&context=amy_hinkler> accessed 16 January 2014.

147 Maria Tzanou, 'Is Data Protection the Same as Privacy? An Analysis of Telecommunications' Metadata Retention Measures' (2013) 17(3) Journal of Internet Law, 28.

148 *ibid* 26.

149 Nissenbaum (n 49).

150 Lilian Edwards, 'Privacy and Data Protection Online: The laws Don't Work?' in Lilian Edwards and Charlotte Waelde (eds), *Law and the Internet* (Hart Publishing 2009) 474–75.

device has taken a picture or video of the third person and/or uploaded it on the Internet. If a violation of privacy takes place, however, the injuries that a person will suffer may constitute a very painful psychological experience. Additionally, if this person tries to seek compensation through a court, the compensation will most probably not be adequate or it will be very difficult to determine the compensation since the damage occurring to a person because of privacy violation is intangible.

Therefore, to create a more complete framework of privacy protection in the above-mentioned situation, the legal framework that is recommended above for the USA or the one that exist in the EU, especially after the homogenization that the Regulation will accomplish, has to be complemented by technical and organizational solutions connected to social aspects, to create an *a priori* protection that will help minimize the violations of privacy.

5.2.1 Controlled systems

As already referred to above, one concern connected to wearable AR devices, such as Google Glass, is that filming and photographing takes place seamlessly and without third persons being able to notice. In other words, the action of the user of the device is not obvious and visible to the people around, making them feel uneasy and uncomfortable, since they believe that they are being monitored all the time. Therefore, a solution that would make this lack of visibility disappear could minimize the discomfort that third persons feel when they are near a user of a wearable AR device.¹⁵¹

As an example of a way to minimize this discomfort, we can think of the CCTVs: to reduce the discomfort that individuals feel knowing that CCTV systems are installed on the streets or in the premises of a company, campus, etc., signs that inform that an area is monitored must be placed in visible places according to laws.¹⁵² In this way, the law asks from the private entities to create appropriate policies, for example, create and place these signs, to succeed transparency and visibility.

In the same way, visibility should be ensured when people use wearable AR devices. In this situation, though, technology will ensure visibility. Specifically the devices should, by some method, inform the third persons around them, making the 'socially significant information visible',¹⁵³ namely that a photo or video is taken at a specific moment. This could be achieved either by some sound or light that

151 See David Pogue, 'Why Google Glass Is Creepy: Scientific American' (21 May 2013), Scientific American <<http://www.scientificamerican.com/article.cfm?id=why-google-glass-is-creepy>> accessed 11 January 2014; and Alexei Oreskovic, 'Google Glass: Cool or Creepy?' (2013), San Jose Mercury News <http://www.mercurynews.com/business/ci_23274277/google-glass-cool-or-creepy> accessed 5 December 2013.

152 For example, Information Commissioner's Office, 'CCTV Code of Practice' Revised edition 2008 <http://www.ico.org.uk/for_organisations/data_protection/topic_guides/~media/documents/library/Data_Protection/Detailed_specialist_guides/ICO_CCTVFINAL_2301.pdf> accessed 9 January 2014; and Information and Privacy Commissioner of Ontario, 'Guidelines for the Use of Video Surveillance Cameras in Public Places' (2007) <<http://www.ipc.on.ca/images/Resources/video-e.pdf>> accessed 9 January 2014.

153 Thomas Erickson and Wendy A Kellogg, 'Social Translucence: An Approach to Designing Systems That Support Social Processes' (2000) 7(1) ACM Transactions on Computer-Human Interaction (TOCHI) 61.

makes apparent that the device is operating at that time. Furthermore, this sign will lead also to the user's awareness, namely the user will be able to see who feels discomfort/does not consent to the user's action, and so the user will know when to stop his/her action. In addition, if the user continues to take pictures and videos of persons who did not give their consent and these photos are uploaded on the Internet, then the legal framework based on the data protection approach that was described above will hold the user accountable for his/her action.

The companies that produce these devices and provide them with an operating system will have to make sure that no applications can surpass these privacy policies, maintaining this way the visibility of the action. As an example, the case of Microsoft and Xbox could be used, even though Microsoft's policy and actions are related to intellectual property protection and not privacy: Microsoft mentions in the policies of Xbox Live¹⁵⁴ that if the terms of use of Xbox Live are violated, Microsoft can suspend permanently a console from connecting to Xbox Live, namely from connecting to the Internet. One of the reasons that a console can be suspended/banned is that the user of the console physically opens and modifies the console to play illegitimate software, such as illegally downloaded games.

Now in the case of the wearable AR device, it is recommended that companies producing the devices and their operating systems produce them in a way that makes it visible and obvious when the user of a wearable AR device is taking a photo or a video. Furthermore, the terms of use of these devices must prohibit the surpassing of the above-mentioned measure. In case of surpassing, penalties should be awarded, such as the ban of the device from the Internet.

In this way, a person will be able see if the user of a wearable AR device is taking his/her photos or videos, and therefore he/she will be able to give or deny his/her consent for that action in the same way that a person now can see if somebody with a camera is photographing him/her.

There is of course a possibility that some users will violate the company's policy and surpass the technical measure anyway. The number, however, of those who are willing to violate the terms of use of technological devices is not very high, since these devices are expensive—for example, Google Glass costs now \$1500—and therefore there are not many users who want to risk a possible incapacitation of such a device.¹⁵⁵

Another problem, however, is that even if a third person takes notice of the fact that he/she is being photographed or videotaped by a user of a wearable AR device, it is possible that he/she will not have the time to react. Specifically, since the wearable AR devices are always connected to the Internet and they have the ability to upload in near-real-time photos and videos on SNS platforms, it is possible that the photo or video of a third person is uploaded on the Internet before he/she has any

154 'Xbox Live Policies on Console Suspensions and Bans', Microsoft <<http://enforcement.xbox.com/Overview/Xbox-Live-Console-Bans>> accessed 9 January 2014.

155 See also Justin Mann 'Xbox 360 Bans Rise to 1 Million, Banned Consoles Go up for Sale' (12 November 2009), TechSpot <<http://www.techspot.com/news/36949-xbox-360-bans-rise-to-1-million-banned-consoles-go-up-for-sale.html>> accessed 11 January 2014, where it is stated that until 2009 only 5 per cent of Xbox users had violated the terms of use of Xbox in question.

time to react. To address this problem and create a real *a priori* protection of privacy, the following solution is recommended.

5.2.2 The face recognition as a solution

As a more radical solution that combines technology, company policies, social norms and the law, face recognition technology, which by many people, as mentioned above, is regarded as a privacy intrusive technology, is hereinafter proposed as a solution to protect privacy from violations by the use of wearable AR devices.

This recommendation is based on the following facts:

- i. Most people have one account on at least one SNS¹⁵⁶ and many SNSs already use face recognition technology.¹⁵⁷
- ii. To upload something on an SNS, the user of a wearable AR device must have an account. Furthermore, to make use of the wearable AR device, an account connected to the provider of the operating system is required.¹⁵⁸ Therefore, the user of a wearable AR device that uploads something on the Internet is traceable and identifiable.
- iii. A hypothesis that people in the near future will consent to face recognition from the SNSs, is taken into consideration. The hypothesis is based on the fact that people are eager to give up some of their privacy to take something back as a benefit¹⁵⁹; at the same time SNSs know that biometrics is a big pool of personal data that they can monetize. Therefore, we can conclude with relative certainty that the SNSs will try to find ways to 'reward' their users to get the data that can be extracted by face recognition. Even the solution that is recommended here could be used as a 'reward' for the users of the SNSs, namely to consent to face recognition to protect privacy, as it will be explained further down.

The main concept of the recommended solution here is that the ones who want to *a priori* protect their privacy against possible violation by other individuals will give permission to the SNSs to use face recognition software. The SNSs, which will now have the data needed for face recognition, shall become the guardian of privacy for the individuals by checking constantly if anyone has uploaded a video or photo that includes a user of the SNS.

Specifically, if a third person is being filmed or photographed and this photo or video is uploaded on an SNSs, such as Facebook or YouTube, then the SNS will proceed to face recognition. If a face is detected, the SNS will first blur the face and

156 See 'Number of Active Users at Facebook over the Years' (2013), Yahoo News <<http://news.yahoo.com/number-active-users-facebook-over-230449748.html>> accessed 16 January 2014.

157 See Yana Welinder, 'A Face tells more than a Thousand Posts: developing face recognition privacy in social networks' (2012) 26(1) Harvard Journal of Law & Technology 165.

158 As an example taken from smartphones, to use an Android device, the user needs a Google account.

159 'Is Online Privacy over? Findings from the USC Annenberg Center for the Digital Future Show Millennials Embrace a New Online Reality' (22 April 2013), University of South California Annenberg, School for Communication and Journalism <http://annenberg.usc.edu/News%20and%20Events/News/130422CDF_Millennials.aspx> accessed 16 December 2013.

when the face is recognized, the SNS will send a notification to the account of the recognized person in which it will state who uploaded the photo and it will ask for permission to 'unblur' the face.¹⁶⁰ In this way, every person has control of his/her data that are uploaded on the Internet and the data will be used in the context that he/she really wants. It has to be mentioned here that there are, however, questions related to interoperability of the SNSs' platforms which have to be studied and analysed further. For instance, what will happen in the case when a third person's photo or video is uploaded on an SNS where this person does not have an account and cannot be contacted?

In this recommended solution, visibility, awareness and accountability are also secured: the person whose face was recognized in an uploaded photo or video gets a signal/message, thus making visible the action of uploading; the person that uploaded the photo or video is aware that another person has been recognized in this photo or video and also has been informed of that. This will lead to a change in the behaviour of the users of wearable AR devices and SNSs, since when they know that they can be held accountable for their actions they will be more careful in what they upload.

It could be argued that this technical solution creates obstacles to the right of freedom of expression since the uploaded photos or videos will have blurred faces until consent is given. Even though it is hard to specify what the right to freedom of expression is and what the limits to this right are, it should be mentioned that it is not an absolute right. That means that it does not prevail over all other rights. Since it is not the scope of this article to analyse how the right to freedom of expression is interacting with the right to privacy, I will only mention that until now SNSs in their terms of use include terms that limit the right to freedom of expression, such as in occasions of vulgar photographs or speech of hatred, etc. In the same manner, the policies of the SNSs could be able to limit the right to freedom of expression to the point of blurring the face of a third person on a photo or video.

6. CONCLUSIONS

Google Glass, as the first commercial wearable AR device, has managed to create big discussions regarding the use of these devices and privacy. Even though these issues are not new, the seamless and surreptitiously way that these devices work has led to a qualitative and quantitative change of these issues. Individuals are concerned not only about being under surveillance by governmental authorities, but also about being under 'sousveillance' by every individual all the time.

The need to create a legal, technological and social framework that will address these issues will be increasingly necessary. Therefore, recommendations as the above mentioned should be taken into consideration by all stakeholders, such as legislators, companies and individuals, so that useful technologies like the wearable AR devices are further improved without becoming a threat to individuals' rights.

160 See n 74, 6.